

Ky–Fan Energy Loss and a Direct Reset Packet Atlas

A Gauge-Free Exit from the Recursive Direction Wall

Prime Dynamics Theory Program

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Abstract

RH-150 proved that source-aligned packet balls are initially tiny but that a separated cross-direction–Ritz radius recursion loses all ten RH-96 chains by the third update. We investigate a gauge-free replacement based on captured energy.

For a positive semidefinite matrix A with rank- r top projector P , gap $g = \lambda_r - \lambda_{r+1} > 0$, and any rank- r projector Q , define the Ky–Fan deficit

$$L = \text{tr}(PA) - \text{tr}(QA).$$

We prove the sharp bounds

$$\|P - Q\|^2 \leq \frac{L}{g}, \quad \|P - Q\|_F^2 \leq \frac{2L}{g}.$$

Because every recursive Ritz search space contains its input packet, the captured energy is monotone independently of the threshold branch. This gives a scalar branch-free loss recursion involving only the previous loss, the previous gap, global-packet drift, and Gram drift. The theorem is valid but the finite audit is negative: only 11 of 130 snapshot bounds remain informative as small gaps amplify global-packet motion.

We then reset independently at each source-memory snapshot. The exact binary memory Gram is enclosed around its numerical center with 384-bit Arb arithmetic; a polar-corrected long-double Rayleigh audit supplies the center gap. All 130 native clock-rank packets certify. The minimum ratio of center gap to twice the matrix radius is 60.368, and the largest projector radius is 0.008352. Direct resets also maximize captured energy and therefore dominate every same-rank recursive packet at the same snapshot. This is a finite reset atlas, not an all-level gap theorem and not an identification of the exact threshold-recursive packet. Transition coherence is the next open interface.

1 A sharp energy-to-angle theorem

Let $A \succeq 0$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and let P project onto the first r eigenvectors. For another rank- r orthogonal projector Q , write $q_j = \langle e_j, Qe_j \rangle$ and

$$\theta = r - \text{tr}(PQ) = \sum_{j>r} q_j = \frac{1}{2} \|P - Q\|_F^2.$$

Theorem 1 (Ky-Fan deficit controls packet angle). *If $g = \lambda_r - \lambda_{r+1} > 0$, then*

$$L := \text{tr}(PA) - \text{tr}(QA) \geq g\theta.$$

Consequently

$$\boxed{\|P - Q\| \leq \sqrt{L/g}, \quad \|P - Q\|_F \leq \sqrt{2L/g}.}$$

Both constants are sharp.

Proof. The lost top mass and gained bottom mass both equal θ . Hence

$$\begin{aligned} L &= \sum_{j \leq r} \lambda_j(1 - q_j) - \sum_{j > r} \lambda_j q_j \\ &\geq \lambda_r \theta - \lambda_{r+1} \theta = g\theta. \end{aligned}$$

The Frobenius identity above gives the second bound, while the largest principal-angle sine squared is at most their sum θ . A two-dimensional rank-one rotation with spectrum $(\lambda_r, \lambda_{r+1})$ attains equality [Bhatia, 1997]. $\square$

Unlike a selected cross-direction projector, L is invariant under frame rotations. However, dividing by a small spectral gap is unavoidable: if $g = 0$, arbitrary rotations inside the boundary eigenspace cost zero energy.

2 A branch-free scalar recursion

Let $A_{t-1}, A_t \succeq 0$ have top-rank projectors P_{t-1}, P_t . Suppose a rank- r packet Q_{t-1} has deficit L_{t-1} under A_{t-1} , and an update chooses the best rank- r Ritz packet Q_t in any search space containing $\text{ran } Q_{t-1}$. Then

$$\text{tr}(Q_t A_t) \geq \text{tr}(Q_{t-1} A_t)$$

without knowing which threshold width or direction frame was used.

Define

$$D_t = \text{tr}(P_t A_t) - \text{tr}(P_{t-1} A_t), \quad \Delta_t = \|A_t - A_{t-1}\|_F.$$

Theorem 2 (Branch-free energy transport). *If $g_{t-1} > 0$, then the output deficit obeys*

$$L_t \leq D_t + L_{t-1} + \sqrt{\frac{2L_{t-1}}{g_{t-1}}} \Delta_t.$$

Proof. Insert and subtract A_{t-1} in $\text{tr}((P_{t-1} - Q_{t-1})A_t)$. The old term is L_{t-1} , and Cauchy-Schwarz plus Theorem 1 bounds the drift term by $\|P_{t-1} - Q_{t-1}\|_F \Delta_t$. The Ritz monotonicity inequality then proves the claim. $\square$

The estimate is structurally attractive but pessimistic. It treats all Gram motion as aligned with the packet error and then divides by the next gap. The audit starts at $L_0 = 0$ and iterates the bound. Only 11 of 130 snapshots retain $L_t < g_t$. Thus removing branches alone does not remove cumulative loss.

3 Independent spectral resets

Accumulation disappears if each packet is defined natively at its own snapshot. Let B_t be the archived binary center, let $\tilde{A}_t$ be the exact binary memory Gram, and suppose

$$\|\tilde{A}_t - B_t\| \leq \delta_t.$$

Theorem 3 (Direct reset packet). *If the rank- r center gap g_t satisfies $g_t > 2\delta_t$, the exact top packet $\tilde{P}_t$ and center packet P_t obey*

$$\|\tilde{P}_t - P_t\| \leq \frac{\delta_t}{g_t - \delta_t}.$$

The non-crossing gate is sharp from a norm ball alone.

Proof. This is the Davis–Kahan approximate-gap theorem [Davis and Kahan, 1970, Stewart and Sun, 1990]. If $g_t \leq 2\delta_t$, a diagonal two-mode perturbation can close or reverse the boundary gap. $\square$

The reset packet also has a variational advantage:

$$\text{tr}(P_t B_t) \geq \text{tr}(Q B_t)$$

for every rank- r packet Q . Thus replacing an RH-96 recursive packet by the reset packet cannot worsen the frozen captured-energy tail. It changes the construction, but it does so in the favorable direction for all Ky–Fan tail gates.

4 The 130-snapshot atlas

For each of five scales and two channels, the audit rebuilds all memory Grams from the exact binary operator and source. The chain lengths are 5, 7, 12, 18, 23 snapshots per channel, totaling 130. Arb evaluates the state, trace normalization, and memory recurrence at 384 bits. The maximum matrix operator radius is 2.642×10^{-15} .

Computing 130 full interval eigendecompositions at dimensions up to 256 would be unnecessarily expensive. We instead diagonalize the center, recompute its Rayleigh matrix in long double, bound the off-diagonal Frobenius norm, and correct the almost-orthogonal eigenframe by its polar metric defect. The resulting center spectral error is subtracted twice from the displayed gap; the Arb matrix radius is then inserted into Theorem 3. This is a hybrid outward audit, not a formal interval eigensolver archive [Johansson, 2017].

All 130 packets certify. The smallest gap lower is 1.220×10^{-13} , but the corresponding matrix ball is smaller still; the global minimum $g_t/(2\delta_t)$ is 60.368. The maximum packet radius is 0.008352.

For comparison, the actual frozen recursive packets have informative Ky–Fan angle bounds on 114 snapshots, but their maximum direct distance from the global packet is essentially one. The analytic Ky–Fan inequality has zero dominance failures after outward center-error padding. Independent resets dominate recursive captured energy on all 130 snapshots.

quantity	0.16	0.08	0.04	0.02	0.01
snapshots, both sides	10	14	24	36	46
direct reset certificates	10	14	24	36	46
worst reset radius	6.93×10^{-4}	2.81×10^{-6}	4.54×10^{-5}	6.30×10^{-3}	8.35×10^{-3}

Table 1: Independent clock-rank reset packets.

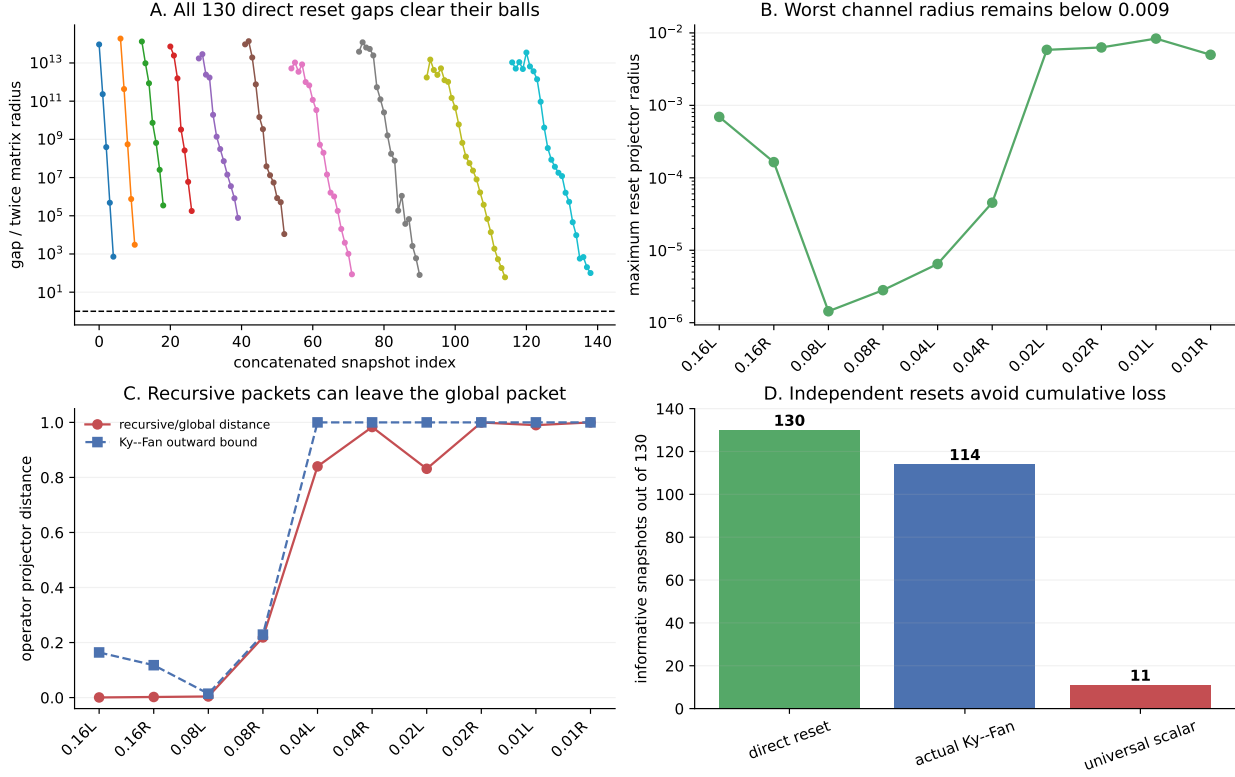


Figure 1: Direct reset gaps and radii, recursive/global packet separation, and the three information counts.

5 Consequence and boundary

RH-150's wall is not a source-gap failure. It is a failure of transporting a specific threshold-recursive gauge through weak selected directions. The same exact binary source model supports a tight top-packet ball at every frozen time when the packet is reset independently.

The next question is geometric rather than spectral existence. Consecutive reset packets must be compared through principal angles, polar Procrustes maps, and pulled-back Grams. If their overlap maps remain uniformly invertible under the certified balls, the reset atlas can replace the failed recursive chain in the outward assembly. If not, a cluster or rank-adaptive packet will be required.

We have not identified reset packets with the exact RH-96 recursive output, inserted them into RH-138, proved a uniform all-level reset gap or transition lower, closed Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

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